

Question Paper

Exam Date & Time: 25-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

Generative AI and Applications [IT_4451]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) A startup wants to create an AI system that can generate customized marketing images for clients with minimal human intervention. They have a small dataset of brand-related visuals and limited computational resources. Identify and justify which type of generative model (e.g., VAE, GAN, or Diffusion Model) you would select for this task. Explain your choice in terms of training stability, data efficiency, and output quality. (5)
- A)

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- B) Using the chain rule of probability, derive the formula for computing the joint probability of a sequence of words in a sentence, and explain how this relates to the working of an RNN language model. (3)

Course Outcome: "CO2" / Cognitive Level: "Application"

- C) Describe how the hidden state h_t and the output prediction $\hat{y}_t$ are computed in an RNN language model. Mention the roles of matrices U , W , and V in this process. (2)

Course Outcome: "CO3" / Cognitive Level: "Understand"

- 2) You are an engineer designing two distinct deep learning models for sequence processing, and you must select the appropriate Recurrent Neural Network (RNN) architecture (based on cardinality) for each task. (5)
- A)

Model A: Real-Time Hand Gesture Recognition

- Goal: The model receives a continuous video stream (a sequence of frames) of a person performing a hand gesture (e.g., "swipe left" or "point"). It must output a single class label ("Swipe," "Point," or "Other") only after the entire gesture sequence is complete.

Model B: Automated Medical Transcription

- Goal: The model receives an audio sequence of a doctor dictating a diagnosis. It must output the written text (a sequence of words) that corresponds exactly to the spoken audio sequence

For each model (A and B), perform the following

- Identify the appropriate RNN Cardinality Type (e.g., Many-to-One, One-to-Many, etc.).
- Justify your choice by clearly explaining how the input and output dimensions match the required task flow.

Identify one key architectural difference between the two models you selected (A and B) that handles the input or output sequence length.

Course Outcome: "CO2" / Cognitive Level: "Analysis"

- B) During the implementation of a Transformer model for text summarization project you are designing, your group notices that the model performs poorly when the word order in the input sentence changes. Suppose the Transformer model is summarizing the following two sentences: (3)
1. "The teacher praised the student after the exam."
2. "After the exam, the teacher praised the student."
Although both sentences convey the same meaning, the model produces very different summaries such as:

- For (1): "Student appreciated by teacher."
- For (2): "Teacher praised after exam event"

Apply your understanding of input embeddings and positional encoding to explain how you would modify the model so that it effectively captures the sequence order information of words.

Course Outcome: "CO3" / Cognitive Level: "Application"

- C) Discuss how Deep Neural Networks differ from Shallow Neural Networks in terms of structure and learning capability. (2)

Course Outcome: "CO4" / Cognitive Level: "Understand"

- 3) Construct and illustrate the Encoder workflow in the Transformer architecture using a neat, labeled diagram. Analyze how components such as Positional Encoding, Self-Attention, Add & Normalize, and Feed Forward layers interact to process input sequences effectively. (5)
- A)

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- B) A decoder-based language model assigns the following probabilities to a 4-token sequence: (3)

$$P = [0.4, 0.3, 0.15, 0.15]$$

Using the perplexity formula

$$PP = 2^{-\frac{1}{N} \sum_{i=1}^N \log_2 P(w_i)}$$

Calculate the model's perplexity. Interpret whether a lower or higher perplexity indicates better performance.

Course Outcome: "CO4" / Cognitive Level: "Application"

- C) List and describe the three main Human Evaluation Approaches used to assess model performance, highlighting the focus of each method. (2)

Course Outcome: "CO5" / Cognitive Level: "Remember"

- 4) Develop a strategy to mitigate ethical and privacy risks during LLM deployment. Discuss at least two mechanisms and explain how they reduce harm while maintaining performance. (5)
- A)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- B) Develop and examine the architecture of a Generative Adversarial Network (GAN). Explain the mathematical model and interaction between the generator and discriminator and discuss how adversarial training enables GANs to produce realistic synthetic data. (3)

Course Outcome: "CO5" / Cognitive Level: "Application"

- C) Summarize why transparency and explainability are important for building user trust in AI systems based on large language models. (2)

Course Outcome: "CO1" / Cognitive Level: "Understand"

- 5) Develop and interpret the working mechanism of the Decoder in Transformer architecture. Examine the roles of Masked Self-Attention, Encoder-Decoder Attention, and Feed Forward layers in generating coherent output sequences, highlighting how each contributes to effective language modeling. (5)
- A)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

- B) During fine-tuning of a Transformer model, the decoder's outputs seem semantically inconsistent and lack fluency, even though attention layers are functioning properly. *Utilize your knowledge of the feed-forward network and residual connections in the decoder to suggest how they enhance representation learning and stabilize training.* (3)

Course Outcome: "CO1" / Cognitive Level: "Application"

- C) Compare Prompt Engineering and Fine-Tuning based on their purpose and flexibility in adapting language models to different tasks. (2)

Course Outcome: "CO2" / Cognitive Level: "Remember"

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Event Name	Subject
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058-END SEMSTER-ODD-2025 IT_4451:Generative AI and Applications

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	10
4	4	CO 4	10
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-END SEMSTER-ODD-2025 IT_4451:Generative AI and Applications

#	Level	No of Questions	Marks (%)
1	Remember	2	4(8%)
2	Understand	3	6(12%)
3	Application	5	15(30%)
4	Analysis	5	25(50%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)